

Multi-Zone Display Layout Engineering Worksheet

Unit 4 — Day 17 Activity Worksheet | Standard: CTE-CTB MN CTB.03.02

Student Name: _____	Date: _____	Station #: _____
Canvas Spec: 1920 x 1080 (16:9)	Preset Name: _____	Peer Reviewer: _____

Section 1: Canvas Spatial Layout Mockup

Assign pixel dimensions and visual content targets for each quadrant on the 1920x1080 LED display.

ZONE A: MAIN LIVE VIDEO FEED Dimensions: _____ x _____ px Aspect Ratio: [☐] 16:9 [☐] 4:3 Source: _____	ZONE C: BRANDING / HEADLINES Dimensions: _____ x _____ px Type: [☐] Static [☐] Graphic Bug Text/Logo: _____
ZONE B: CONTINUOUS LIVE NEWS TICKER (BOTTOM BANNER) Dimensions: 1920 x _____ px Font Spec: Bold Sans-Serif Scroll Speed: _____ px/sec	

Section 2: Technical Zone Specifications Matrix

Display Zone	Target Content & Source	Calculated Pixels	Color & Contrast Specs
Zone A Main Video	Live Anchor Studio Feed / NDI Input	Width: _____ px Height: _____ px	Full Motion Video / Native SDI
Zone B News Ticker	Breaking News Text / RSS Feed	Width: 1920 px Height: _____ px	Font: _____ Bg Color: _____
Zone C Branding & Stats	Station Bug Logo & Live Score Graphics	Width: _____ px Height: _____ px	High Contrast Vector PNG Opacity: 100%

Section 3: Pixel Calculations & Design Verification

1. **Vertical Pixel Sum Check:** Verify that Zone A Height + Zone B Height equals 1080 pixels total.
2. **Legibility Calibration:** Why is pure thin cursive text unsuitable for LED display boards under bright outdoor stadium lights?
3. **Control Studio Transition:** Explain how preset triggers in Daktronics Display Studio prevent broadcast errors during live shows.

Instructor Review: [☐] Canvas Math Valid [☐] Layout Approved	Signature: _____
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